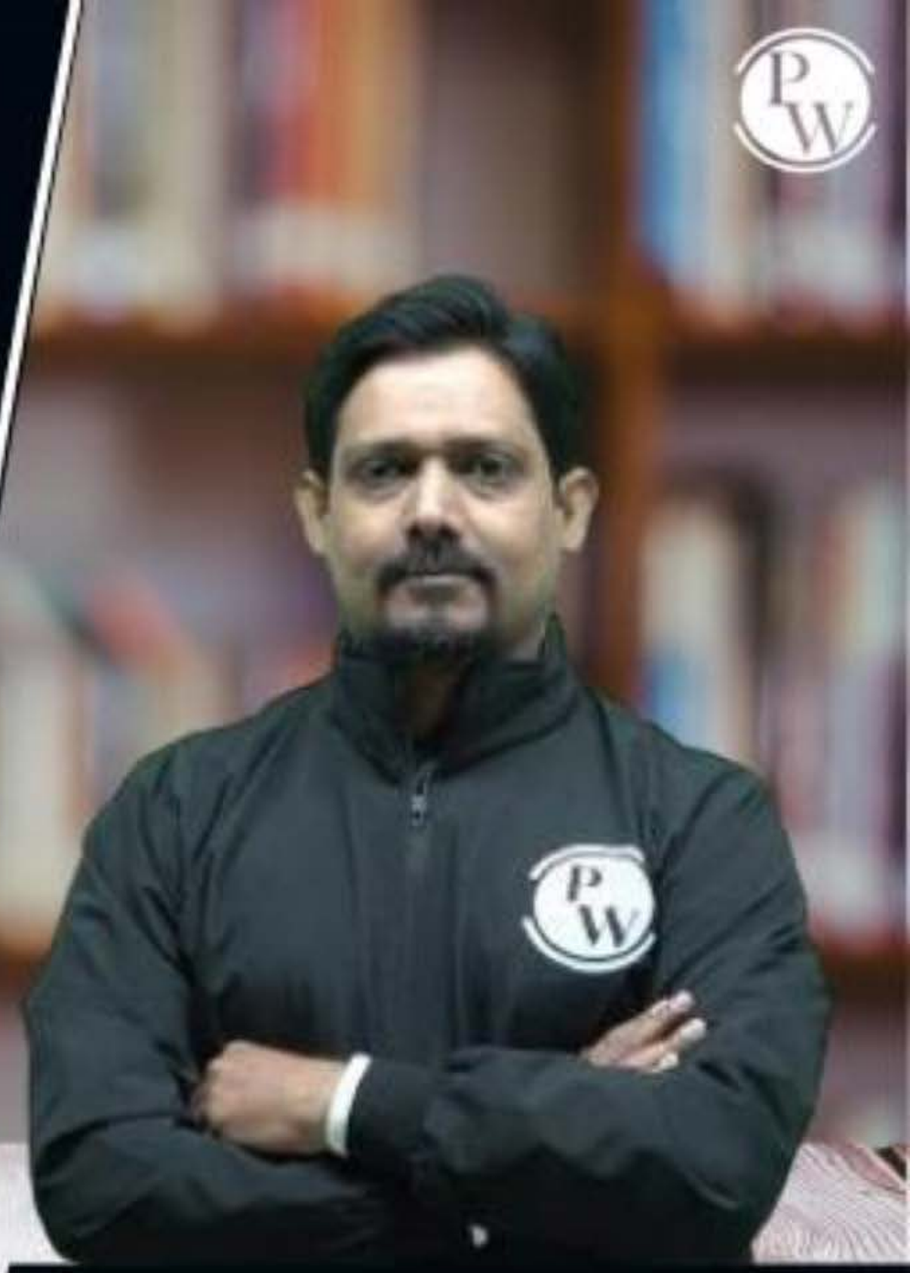
Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 07



By- Abhishek Sir

Recap of Previous Lecture



Topic — AVL tree

Topic — Avl Rotation

Topic — LL, RR

Topic — LR, RL

Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Minimum No. of Node

Heap

Slide



Topic : Tree

AVL Tree Minimum number of nodes

HEAP

Insert in MAX HEAP

Slide



Topic : AVL Tree

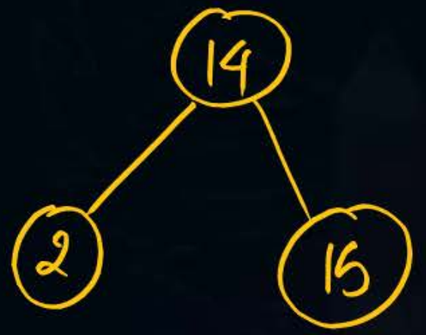
Construct AVL tree of Height h with minimum No. of Nodes

h = 0 — O

Minimum No. of Nodes

h = 1 ——





Slide



Topic : AVL Tree

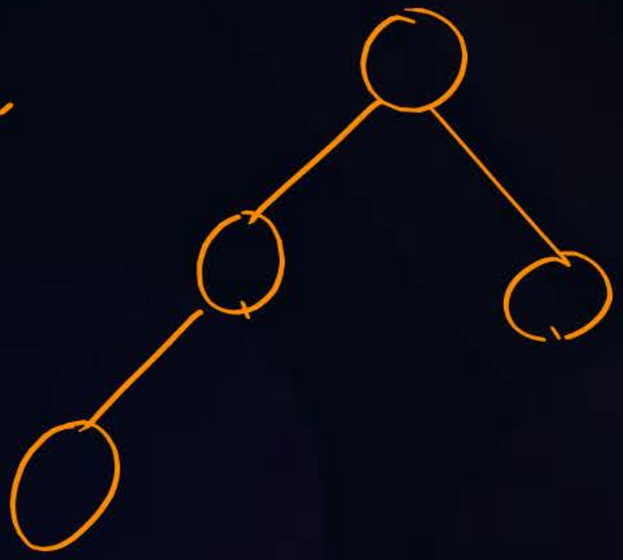
h = 1



N(1) = 2

N(h) represent Minimum No. of Nodes with height h then

h = 2

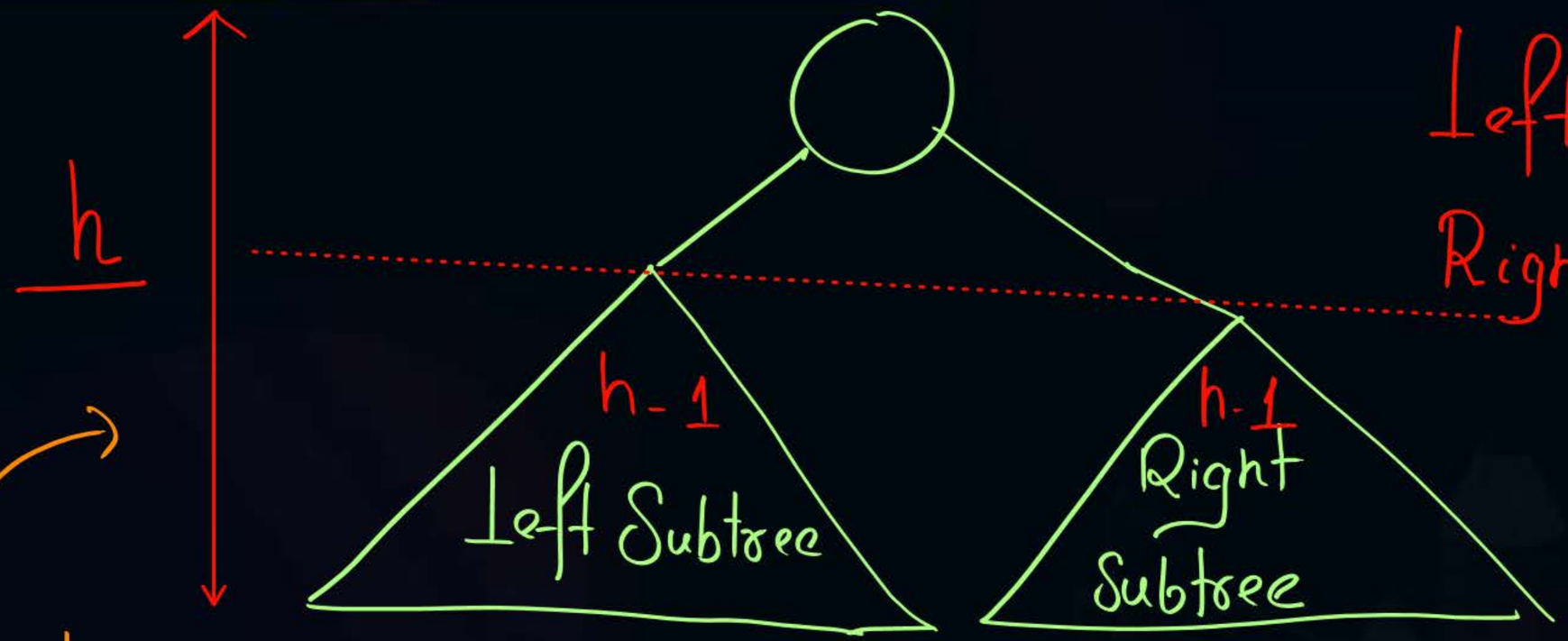


N(2) = 4

Slide



Topic : AVL Tree



Left subtree

Right Subtree

height

$(h-1)$

This will Not result in Minimum No. of Node AVL.

0, 1, -1

Balancing factor

Slide



Topic : AVL Tree

Left & Right subtree's height differen

$(h-1) - (h-1) = 0$

Height difference | 2 - 1 | is also allowed

Slide



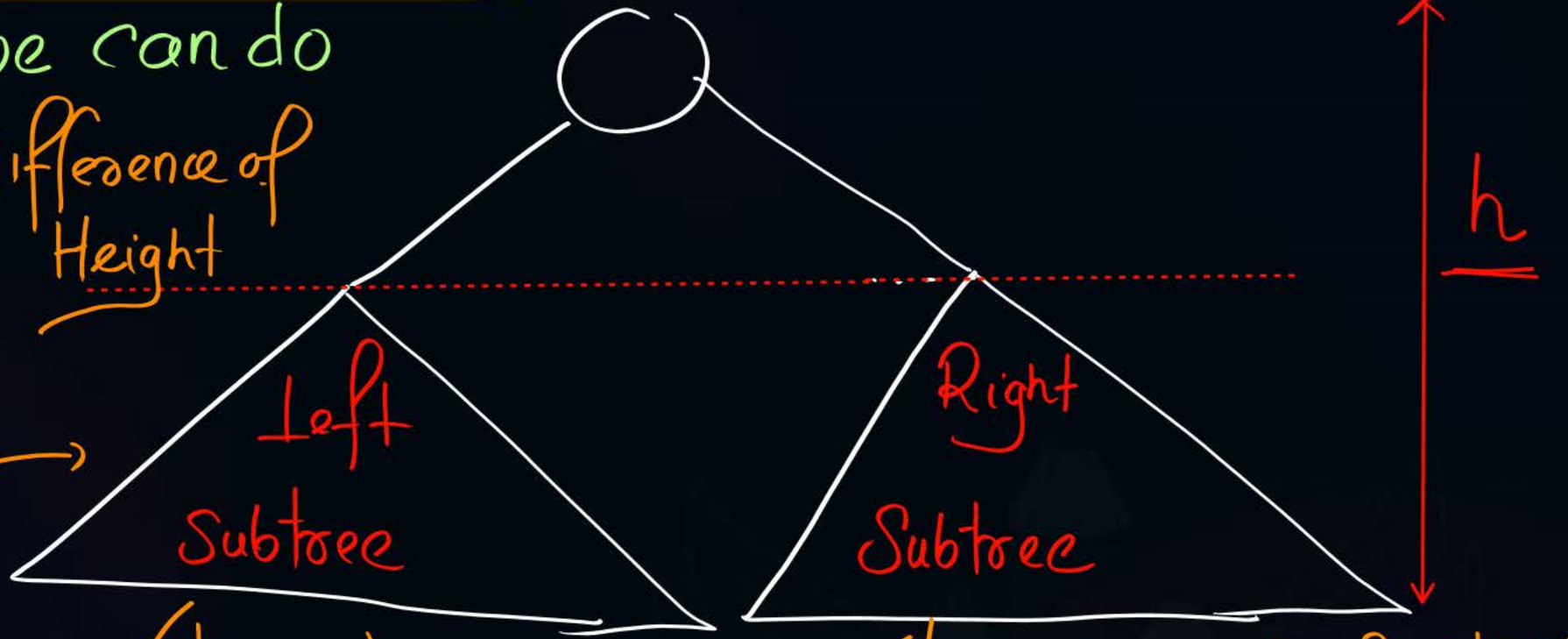
Topic : AVL Tree

What we can do

$(h-1)-(h-2)$ difference of Height

$h-1-h+2$

$= 1$



Left subtree minimum No. Node

$(h-1)$

$(h-2)$

or

$(h-2)$

$(h-1)$

Right Subtree Minimum No. of Nodes

Slide



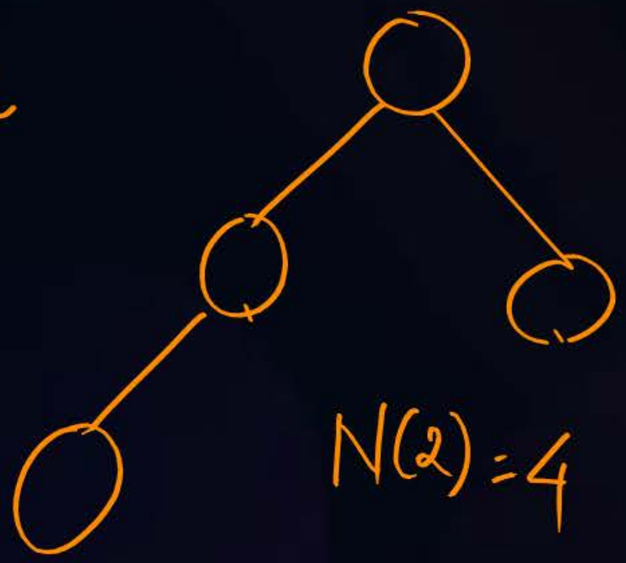
Topic : AVL Tree

h = 1



N(1) = 2

h = 2

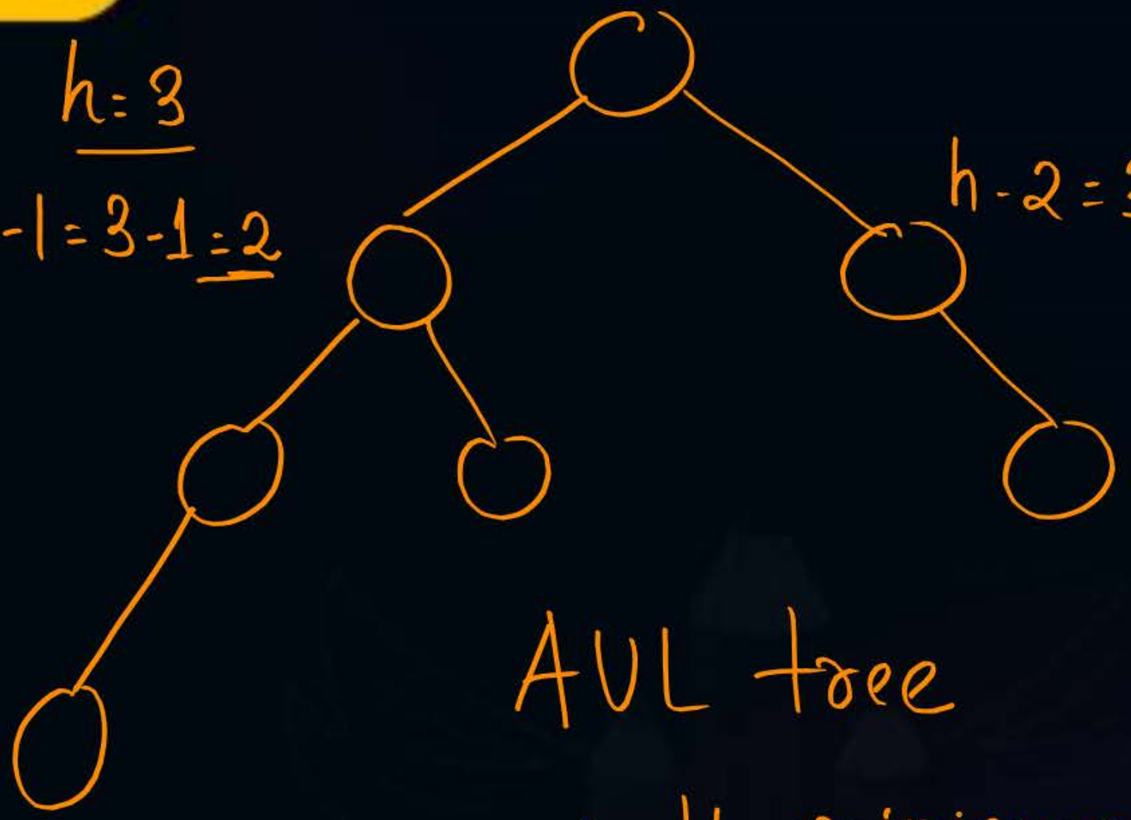


N(2) = 4

h = 3

h-1 = 3-1 = 2

h-2 = 3-2 = 1

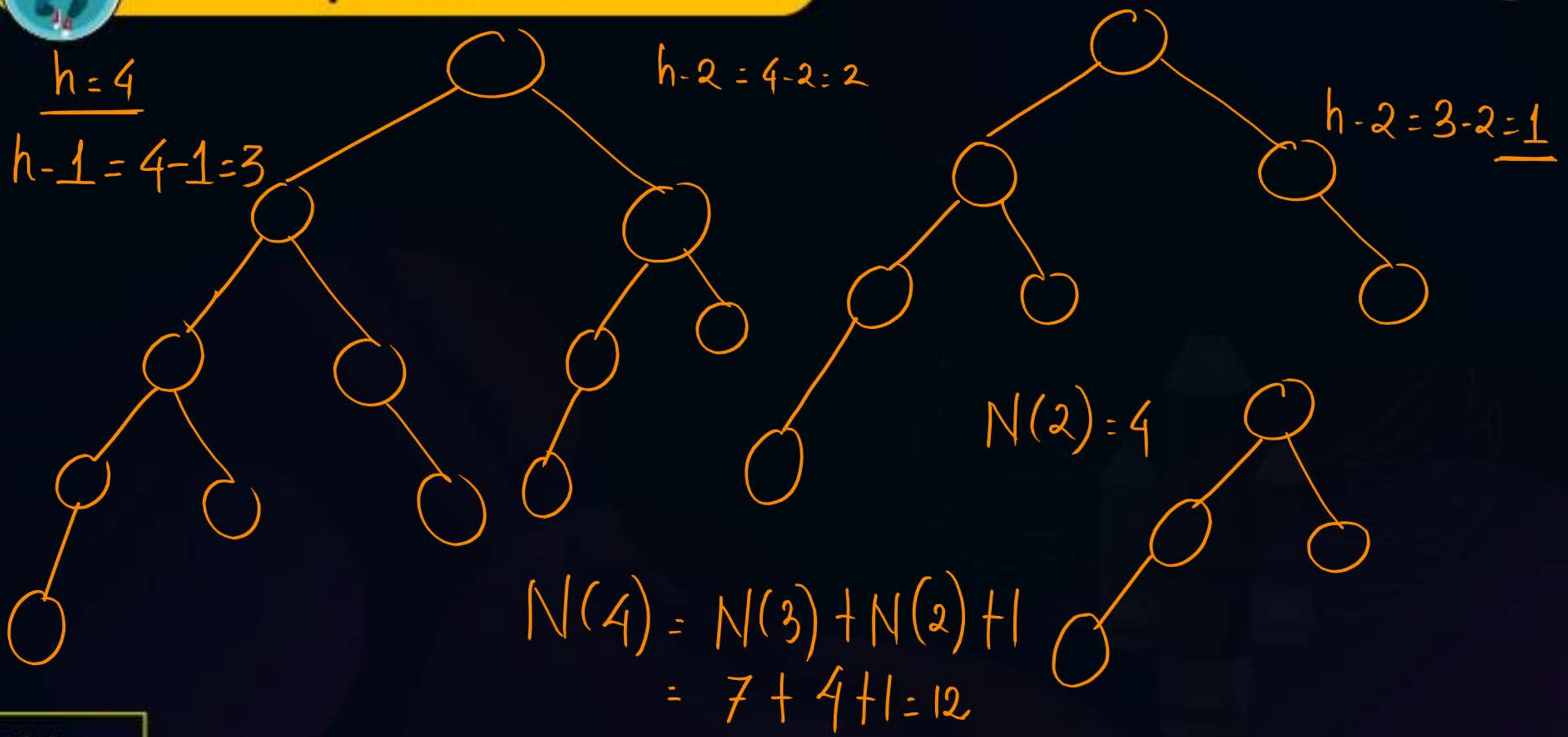


AVL tree with minimum Node at height 3

Slide



Topic : AVL Tree



Slide



Topic : AVL Tree

Minimum No. of Node at height h

$$N(h) = N(h-1) + N(h-2) + 1$$

$$N(0) = 1 \ , \ N(1) = 2$$

Slide



Topic : AVL Tree

What is the minimum No. of Nodes in AVL tree of Height 9 ——— 143

$N(h) = N(h-1) + N(h-2) + 1$

$N(0) = 1$

$N(1) = 2$

$N(2) = 4$

$N(3) = 7$

$N(4) = 12$

$N(5) = 20$

$N(6) = 33$

$N(7) = 54$

$N(8) = 88$

$N(9) = 88 + 54 + 1$

143

Slide



Topic : AVL Tree

Search
Insert
Delete
Min
Max

} Balanced BST

AVL tree

$O(\log_2 n)$

Slide



Topic : AVL Tree

DPP

Height of AVL-tree with minimum No. of Node

$\lceil \log_2 n \rceil$

$\lceil \log_2 2 \rceil = 1$

$\lceil \log_2 4 \rceil = 2$

$\lceil \log_2 7 \rceil = 3$

Slide



Topic : Heap

Heap is a Complete binay Tree

Complete Binay Tree : A Binay tree is called CBT if every level is full except possibly the last.

In last level element will stoictly filled from left to Right.

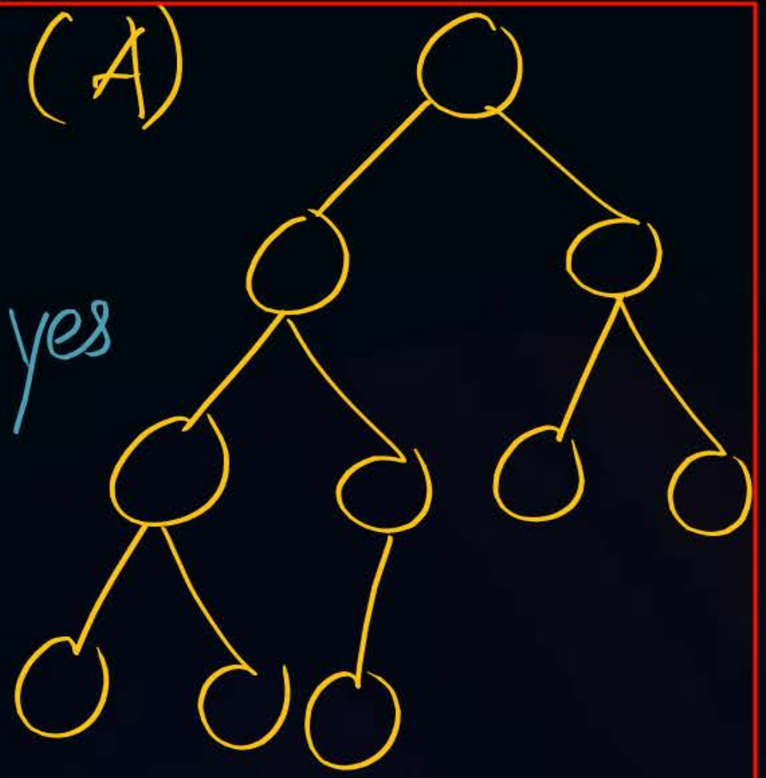
Slide



Topic : Heap

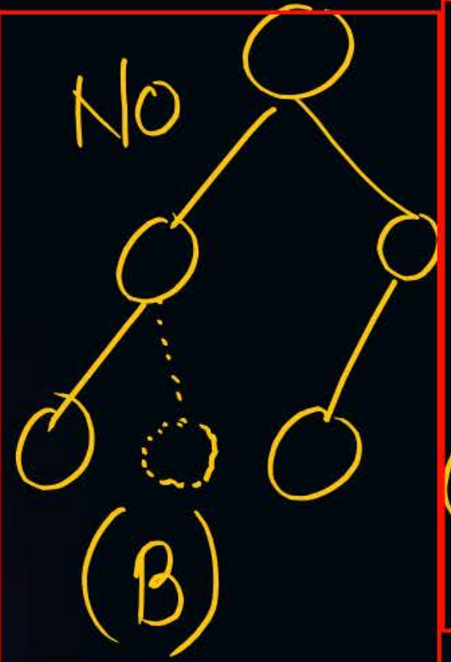
A C D | CBT

(A)

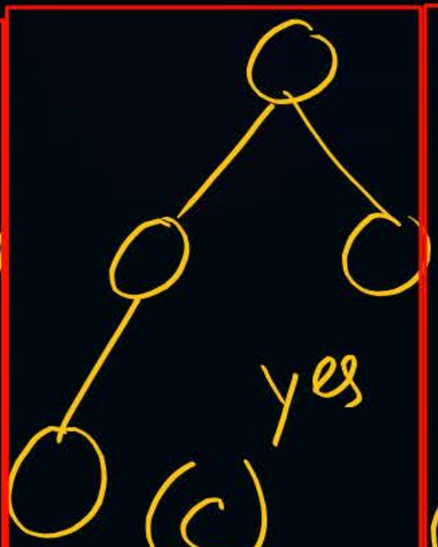


yes

No



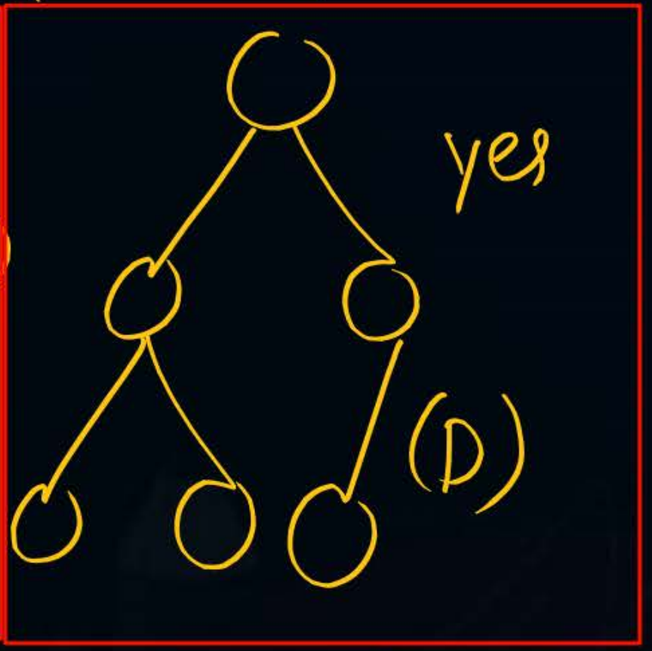
(B)



yes

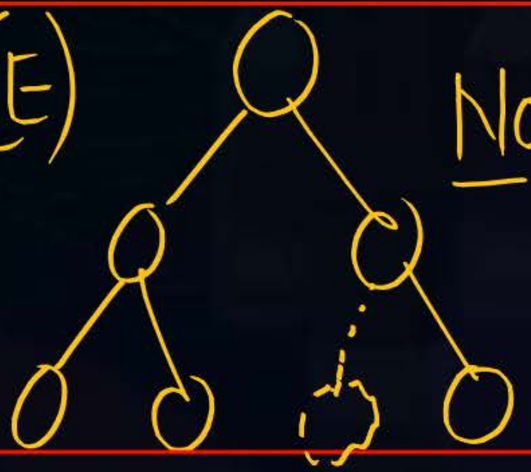
(C)

yes



(D)

(E)



No

Slide



Topic : Heap

if a Heap of Height 4 having maximum No. of Node x and minimum No. of Node as y then x+y is

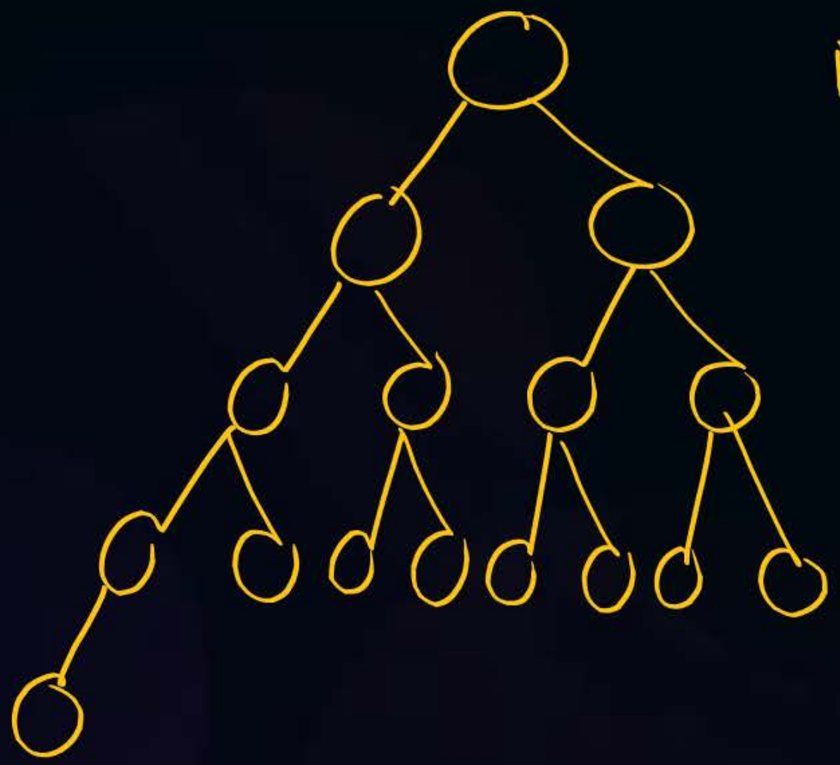
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Slide



Topic : Heap

Max : $2^{h+1} - 1$ $= 2^{4+1} - 1 = 32 - 1 = 31$



Min : 16

$31 + 16 = 47$

Slide



Topic : Heap

Heap is Not Binary Search Tree

Max heap : Max heap is a CBT. In max heap the value at parent Node is greater than value at its child. This property is true for every Subtree.

Slide



Topic : Heap

Sequential Representation of tree is used

Array is used to represent the tree.

if an element at Index

i ($i \neq 1$)

Left child : $2i$

Right child : $2i+1$

parent : $\lfloor i/2 \rfloor$

Slide



Topic : Heap

Max heap by Insertion

1, 4, 7, 2, 3, 5, 6

(I) 1

(II) 1 ⇒ 4 (Swap())

(III) 7

(IV) 2

(V)

(VI) 5

(VII)



Slide



Topic : Heap



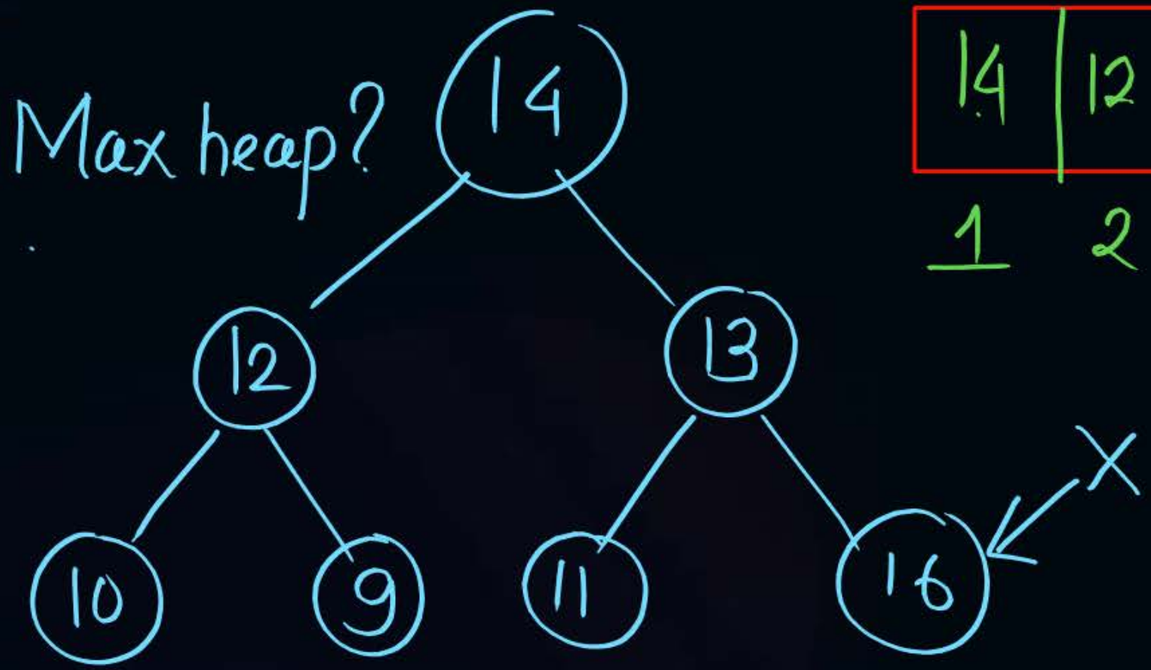
7	3	6	1	2	4	5
1	2	3	4	5	6	

Array Representation

Slide



Topic : Heap



14	12	13	10	9	11	16
1	2	3	4	5	6	7

Slide

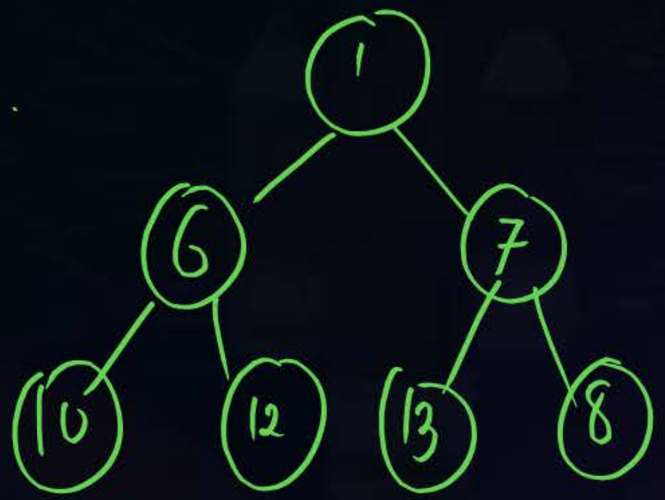


Topic : Heap

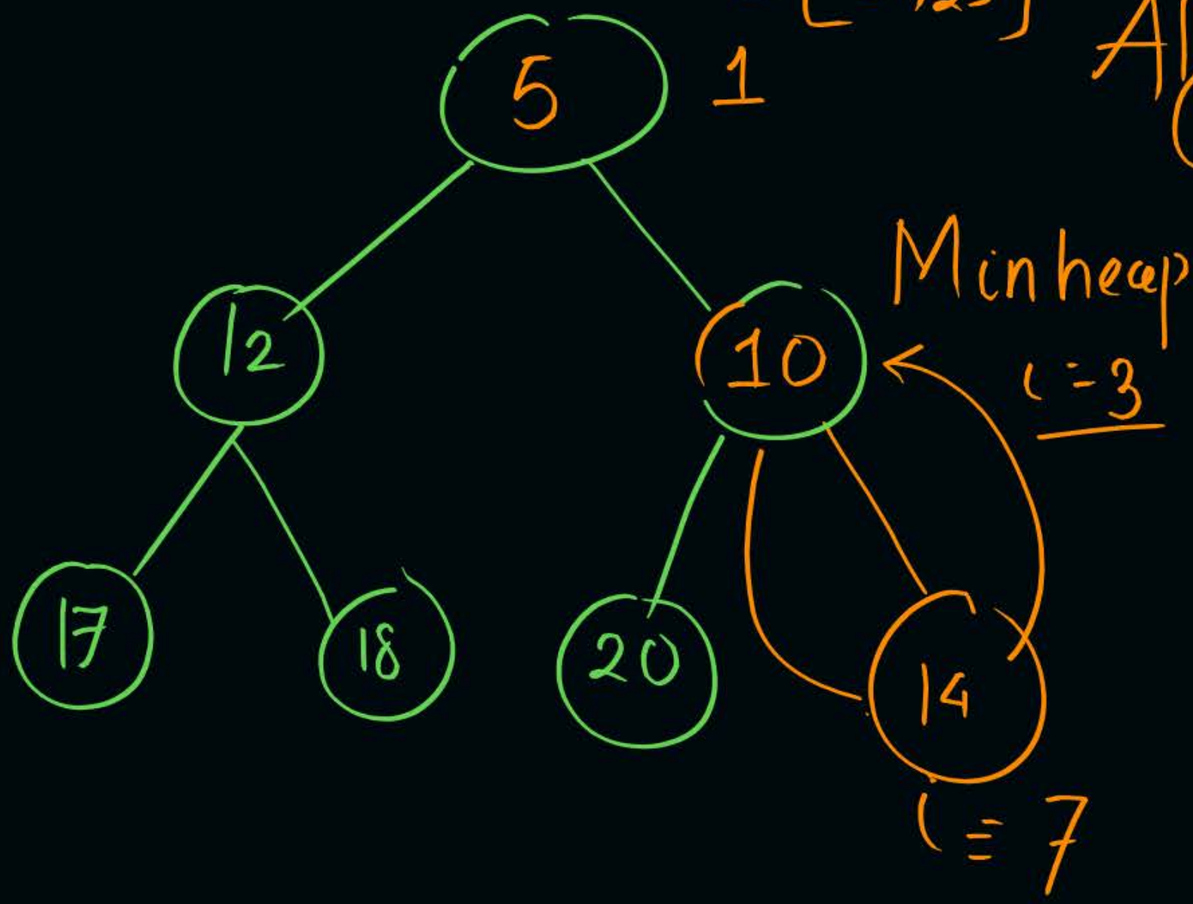
array

1	6	7	10	12	13	8
1	2	3	4	5	6	7

Min heap is CBT, in which the value at parent node is lesser than value at its children. This property is Recursively true for subtree.



Slide




```
Algorithm Insert(a, n, x){
    a[n] = x;
    i = n;
    while(i > 1 and a[⌊i/2⌋] > x){
        Swap(a[i], a[⌊i/2⌋]
        i = ⌊i/2⌋
    }
}
```



Topic : Question

The number of nodes a heap of height k can hold is

(A) 2^k to 2^{k+1}- 1

(B) 2^{k+1} to 2^{k+1}- 1

(C) 2^{k-1} to 2^{k+1}- 1

(D) 2^k -1to 2^{k+1}- 1

Slide

Topic : Heap

Consider a binary max-heap implemented using an array.

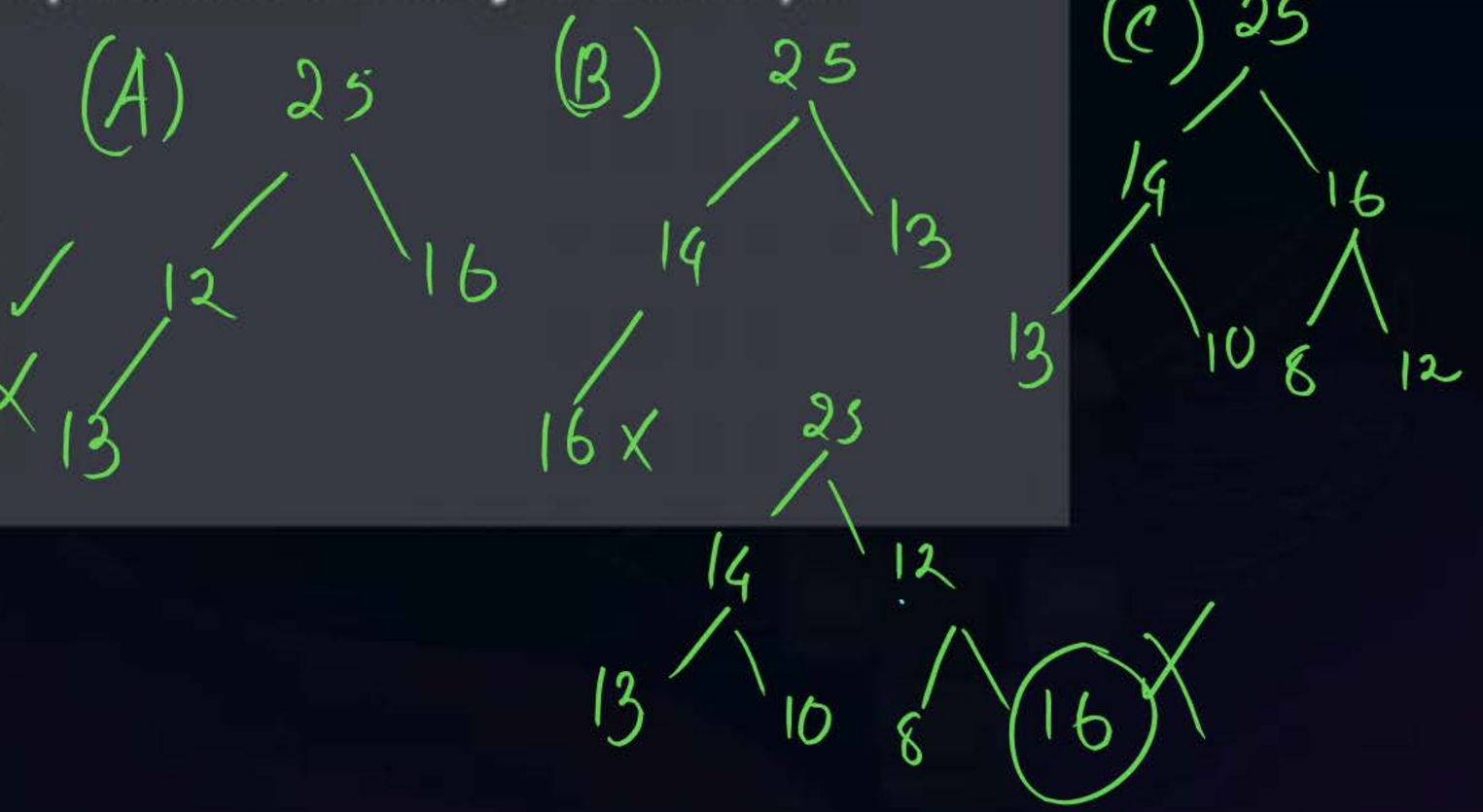
Which one of the following array represents a binary max-heap?

A. $\{25, 12, 16, 13, 10, 8, 14\}$

B. $\{25, 14, 13, 16, 10, 8, 12\}$

C. $\{25, 14, 16, 13, 10, 8, 12\}$

D. $\{25, 14, 12, 13, 10, 8, 16\}$



Slide

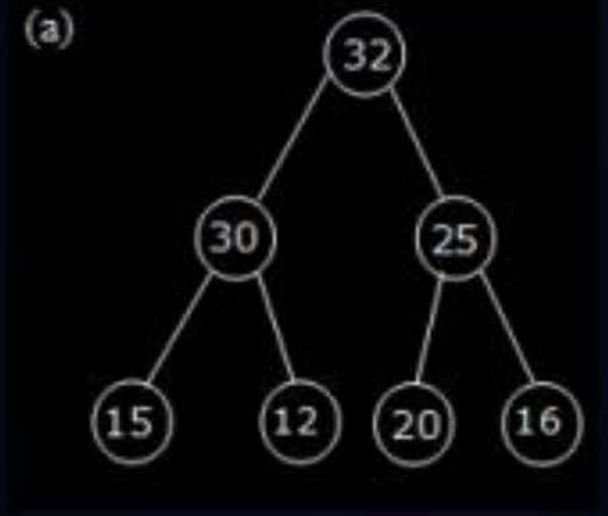


Topic : Heap

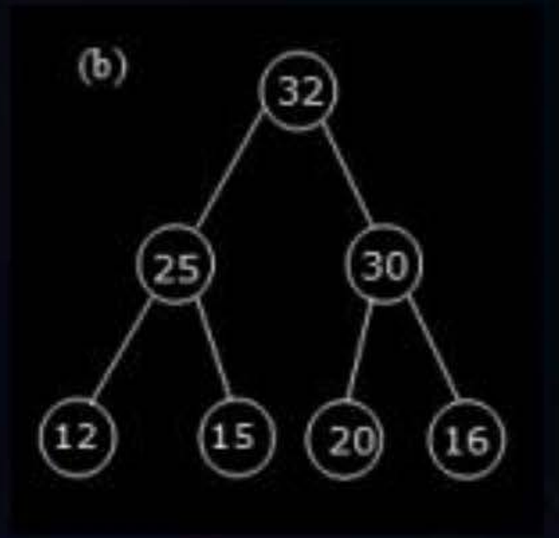
GATE 2004, Question Number 37, 2-Marks,

The elements 32, 15, 20, 30, 12, 25, 16, are inserted one by one in the given order into a maxHeap. The resultant maxHeap is

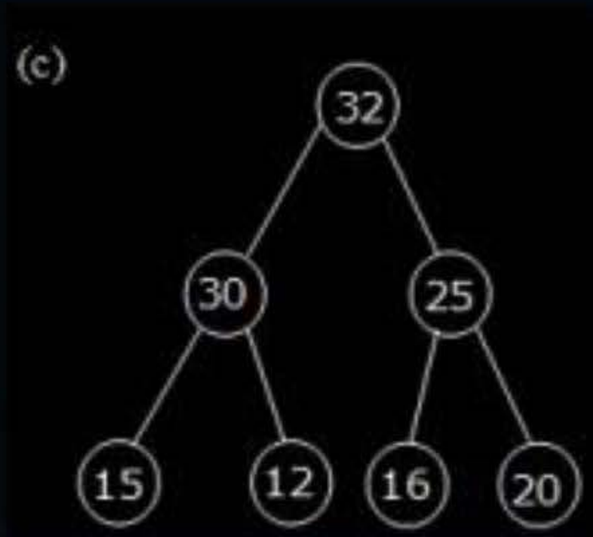
(a)



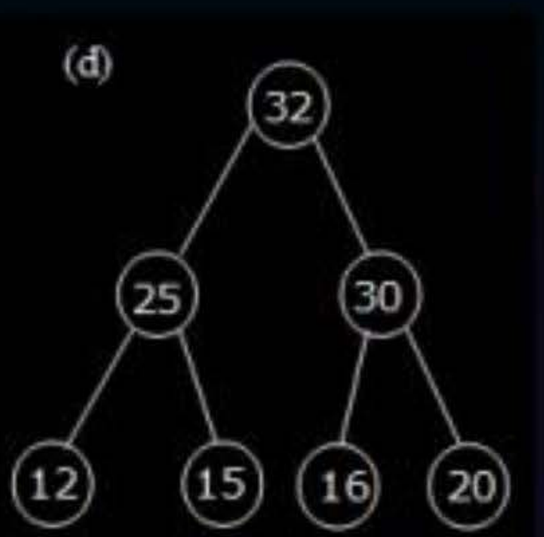
(b)



(c)



(d)

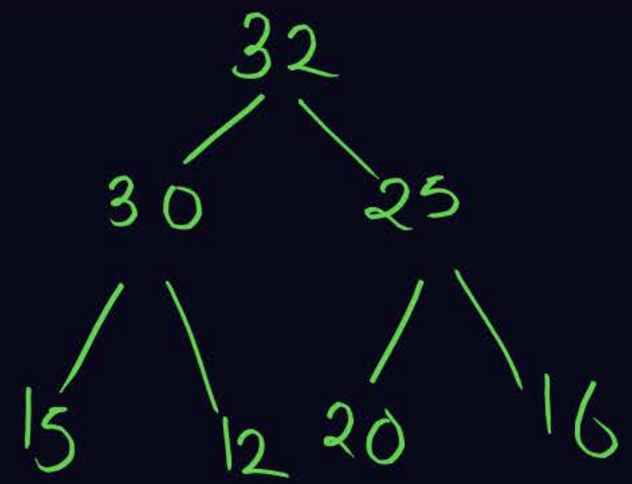




Topic : Heap

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The elements 32, 15, 20, 30, 12, 25, 16, are inserted one by one in the given order into a maxHeap. The resultant maxHeap is



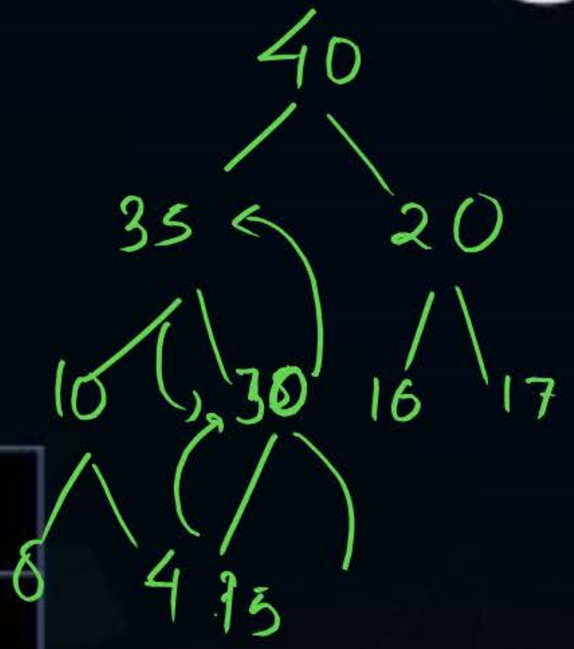
Slide



Topic : Heap

GATE 2015 Set-I, Question Number 27, 2-Marks

Consider a max heap, represented by the array: 40, 30, 20, 10, 15, 16, 17, 8, 4.



Array Index	1	2	3	4	5	6	7	8	9
Value	40	30	20	10	15	16	17	8	4

Now consider that a value 35 is inserted into this heap. After insertion, the new heap is



Topic : Heap

(A) 40, 30, 20, 10, 15, 16, 17, 8, 4, 35

(B) 40, 35, 20, 10, 30, 16, 17, 8, 4, 15

(C) 40, 30, 20, 10, 35, 16, 17, 8, 4, 15

(D) 40, 35, 20, 10, 15, 16, 17, 8, 4, 30

(B)

Slide



Topic : Question

Consider the function and the binary tree shown.

```
typedef struct node {
    int val;
    struct node *left, *right;
} node;
```

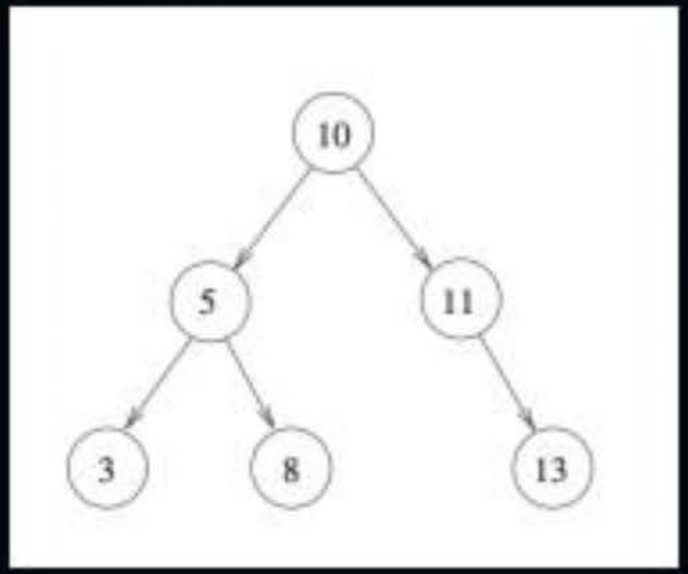
```
int foo(node *p) {
    int retval;
    if (p == NULL)
        return 0;
    else {
        retval = p->val + foo(p->left) + foo(p->right);
        printf("%d ", retval);
        return retval;
    }
}
```

Slide



Topic : Question

Consider the function and the binary tree shown.




```
int foo(node *p) {
    int retval;
    if (p == NULL)
        return 0;
    else {
        retval = p->val + foo(p->left) + foo(p->right);
        printf("%d ", retval);
        return retval;
    }
}
```

A. 3 8 5 13 11 10

B. 3 5 8 10 11 13

C. 3 8 16 13 24 50

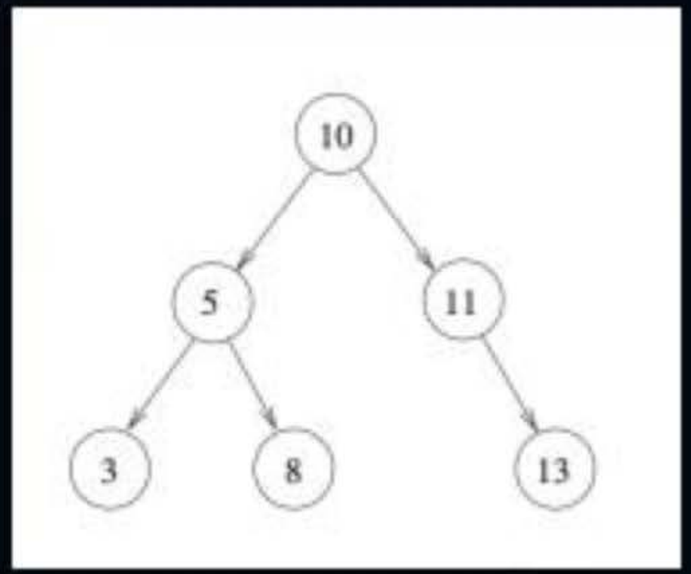
D. 3 16 8 50 24 13

Slide



Topic : Question

Consider the function and the binary tree shown.




```
int foo(node *p) {
    int retval;
    if (p == NULL)
        return 0;
    else {
        retval = p->val + foo(p->left) + foo(p->right);
        printf("%d ", retval);
        return retval;
    }
}
```

Slide



2 mins Summary

Topic — AVL with minimum No. of Nodes of Height

Topic — CBT

Topic — Heap

Topic — Heap Insertion

Topic

Slide

THANK - YOU